

Abhinav Chinnusamy

abhinav.chinnusamy@wisc.edu | cabhinav.com | (608) 658-7885 | Madison, WI 53703

EDUCATION

University of Wisconsin-Madison

Master of Science in Electrical and Computer Engineering

Expected May 2026

Coursework: Power Electronic Circuits, Power Electronics Laboratory,
Solid State Power Conversion, Introduction to Optimization,
Electric Machines, Advanced Computer Control of Machines and Processes

Indian Institute of Technology

Bachelor of Technology in Electrical Engineering

Dharwad, India

Aug 2020 – Apr 2024

RESEARCH EXPERIENCE

Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC)

Madison, WI

Graduate Researcher

Aug 2024 – Present

- Working with Dr. Jinia Roy on pulsed power supplies for high-voltage applications.
- Designed and validated PCBs for Marx Generators, integrating Ti AM263P for control and testing.

Power and Energy Group, IIT Dharwad

Dharwad, India

Undergraduate Researcher

Dec 2022 – June 2024

- Developed a solid-state circuit breaker (800V, 70A) for EVs and DC homes using SiC devices under Dr. Satish Naik.
- Designed PCBs and conducted double pulse tests for GaN-based half-bridge inverters for MMC applications under Dr. Abhijit Kshirsagar.

ENGINEERING EXPERIENCE

Annapurna Labs (Amazon Web Services)

Austin, TX

Hardware Development Engineer Intern

June 2025 – August 2025

- Developed tools to enhance debugging processes for accelerator card diagnostics.
- Analyzed and repaired RMA cards, identifying and resolving hardware bugs.
- Validated high-density capacitors and supported next-gen GPU card bring-up.

PUBLICATIONS

1. **A. Chinnusamy**, H. Baik, and J. Roy, “High Power Density Solid State Marx Generator Based Pulsed Power Driver for Nonlinear Transmission Line,” in *2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)*, Bengaluru, India, 2025, pp. 1–6, doi: 10.1109/ECCE-Asia63110.2025.11112355.
2. D. Dsa, **A. Chinnusamy**, S. N. Banavath, and E. L. Carvalho, “Implementation of Protection Features for a Modular Bidirectional Solid-State Battery Disconnect,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 6, pp. 6001–6010, Nov. 2024.
3. **A. Chinnusamy**, D. Dsa, and S. N. Banavath, “Intelligent Battery Protection System for Electric Vehicle Applications,” in *2024 IEEE 18th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG)*, Poland, June 2024, pp. 1–6.